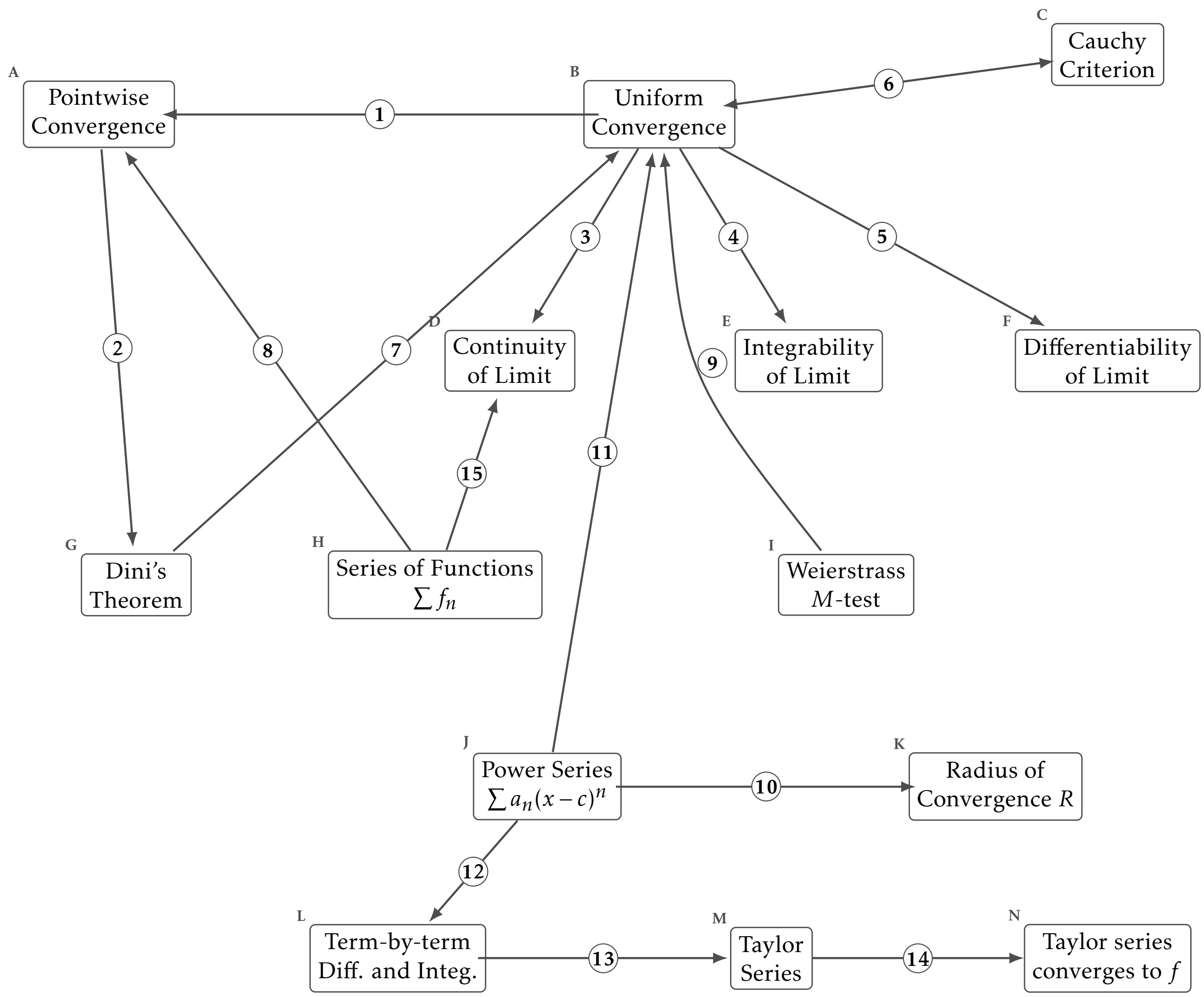
Sequences and series of functions: answers

Math 163 • Prof. Chowdhury

Every arrow, with the reason underneath. Keep this away from the blank worksheet.

What each box means

- **A. Pointwise Convergence.** For each x and each $\varepsilon > 0$ there is an N depending on both, with $|f_n(x) - f(x)| < \varepsilon$ for $n \geq N$.
- **B. Uniform Convergence.** For each $\varepsilon > 0$ there is an N depending only on ε, with $|f_n(x) - f(x)| < \varepsilon$ for $n \geq N$ and every x at once.
- **C. Cauchy Criterion.** $\{f_n\}$ converges uniformly exactly when for each $\varepsilon > 0$ there is an N with $|f_m(x) - f_n(x)| < \varepsilon$ for all $m, n \geq N$ and all x.
- **D. Continuity of Limit.** Each f_n continuous and $f_n \to f$ uniformly gives f continuous.
- **E. Integrability of Limit.** Each f_n integrable and $f_n \to f$ uniformly gives f integrable, with $\lim \int f_n = \int \lim f_n$.
- **F. Differentiability of Limit.** $f_n \to f$ pointwise together with $f_n' \to g$ uniformly gives $f' = g$. It is the derivatives that have to converge uniformly.
- **G. Dini's Theorem.** $f_n \to f$ pointwise on $[a, b]$, monotonely, with every f_n and f continuous, gives $f_n \to f$ uniformly.
- **H. Series of Functions $\sum f_n$.** $\sum f_n$ converges when its partial sums $S_N(x) = \sum_{k \leq N} f_k(x)$ do. A series of functions is a sequence of functions.
- **I. Weierstrass M-test.** If $|f_n(x)| \leq M_n$ for every x and $\sum M_n < \infty$, then $\sum f_n$ converges uniformly and absolutely.
- **J. Power Series $\sum a_n(x - c)^n$.** A series of functions whose nth term is a monomial of degree n centred at c.
- **K. Radius of Convergence R.** $\sum a_n(x - c)^n$ converges absolutely for $|x - c| < R$ and diverges for $|x - c| > R$. The endpoints are checked one at a time.
- **L. Term-by-term Diff. and Integ..** Inside $(c - R, c + R)$ a power series can be differentiated and integrated term by term, and the result has the same radius R.
- **M. Taylor Series.** $\sum f^{(n)}(a)(x-a)^n/n!$. The coefficients are forced by differentiating the power series at its centre.
- **N. Taylor series converges to f.** The Taylor series converges to $f(x)$ exactly when the remainder $R_N(x) = f(x) - P_N(x)$ tends to zero.


A
Pointwise Convergence
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Term-by-term Diff. and Integ.
M
Taylor Series
N
Taylor series converges to f
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2
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The arrows

1. **Uniform convergence is pointwise convergence with an N that does not depend on x, so it is the stronger of the two.** One quantifier moves and everything in the chapter follows from where it lands. Pointwise lets N chase x; uniform does not.

2. **Dini's theorem takes pointwise convergence and three further hypotheses, a closed bounded interval, monotone convergence, and a continuous limit, and concludes uniform convergence.** Pointwise convergence alone preserves neither continuity nor integrability: every x^n on $[0,1]$ is continuous and their pointwise limit is discontinuous at 1. Each of Dini's three hypotheses is necessary, and dropping any one of them admits a counterexample.

3. **A uniform limit of continuous functions is continuous.** The three-epsilon argument: get near f_n uniformly, use the continuity of that one f_n, and come back.

4. **A uniform limit of integrable functions is integrable, and the limit of the integrals is the integral of the limit.** Uniform closeness bounds the difference of the integrals by the length of the interval times ε, which is what lets the two operations swap.

5. **If $f_n \to f$ pointwise and $f_n' \to g$ uniformly, then f is differentiable with $f' = g$, so the hypothesis of uniform convergence falls on the derivatives.** Uniform convergence of f_n alone is insufficient: $\sqrt{x^2+1/n^2} \to |x|$ uniformly and $|x|$ is not differentiable at 0, while $(1/n)\sin(n^2x) \to 0$ uniformly with derivatives $n\cos(n^2x)$ that diverge. After the clean results for continuity and integrability, the situation for differentiability is disappointing.

6. **A sequence of functions converges uniformly exactly when it is uniformly Cauchy, so neither statement needs the limit named in advance.** The criterion never mentions the limit, which is what makes the M-test possible: uniform convergence of a series can be established without writing down its sum.

7. **On a closed bounded interval, monotone pointwise convergence of continuous functions to a continuous limit is already uniform.** Every hypothesis is doing work. Drop compactness, monotonicity, or the continuity of the limit and there is a counterexample waiting.

8. **A series of functions is the sequence of its partial sums, so every statement about sequences of functions applies to it unchanged.** Nothing about series of functions is new. It is the sequence $\{S_N\}$ wearing different notation.

9. **Bounding each term by a constant whose series converges gives uniform and absolute convergence of the series of functions.** Each M_n bounds $|f_n(x)|$ uniformly in x, so $\sum M_n < \infty$ bounds the tails of $\sum f_n$ by a quantity independent of x.

10. **A power series converges absolutely inside one radius and diverges outside it, and that radius is what the ratio or root test computes.** The trichotomy is the theorem. What happens at the two endpoints is decided separately and can go either way.

11. **Inside its radius a power series converges uniformly on every closed bounded subinterval, which is what licenses working on it term by term.** The convergence is uniform on each compact subset of $(c-R, c+R)$ and can fail to be uniform on the whole interval, which is why the statement names a closed bounded subinterval.

12. **Because the convergence is uniform on compact subintervals, a power series can be differentiated and integrated term by term, and the result keeps the same radius.** The interchange theorems supply this, and the radius is unchanged because the ratio or root test sees the same limit for the derived series.

13. **Differentiating term by term at the centre forces the coefficients to be $f^{(n)}(c)/n!$, so a function has at most one power series about a point.** Uniqueness comes free. Any two power series agreeing near a point have the same coefficients, because both sets are computed from the same derivatives.

14. **The Taylor series converges to $f(x)$ exactly when $R_N(x) = f(x) - P_N(x) \to 0$, a condition on f itself.** e^{-1/x^2} extended by 0 has every derivative zero at the origin, so its Taylor series is identically zero and converges everywhere, and it agrees with the function only at 0. The remainder is what decides the question.

15. **Applying the interchange theorems to the partial sums gives the matching statements for the sum of a series of functions.** A uniformly convergent series of continuous functions has a continuous sum and may be integrated term by term, by the sequence theorems applied to $\{S_N\}$.

Claims that look right and are not

- A uniform limit of differentiable functions is differentiable.
- A power series and the series obtained by differentiating it term by term have different radii of convergence.
- A function with derivatives of every order at a point is the sum of its Taylor series near that point.
- Pointwise convergence on a closed bounded interval is automatically uniform.
- A series of continuous functions that converges pointwise has a continuous sum.

Counterexamples and benchmarks

- x^n on $[0,1]$ takes continuous functions to a discontinuous limit, pointwise.
- $\sqrt{x^2+1/n^2} \to |x|$ uniformly, and the limit is not differentiable at 0.
- $(1/n)\sin(n^2x) \to 0$ uniformly while its derivatives diverge.
- $x/(1+n^2x^2) \to 0$ uniformly, and the derivative of the limit is not the limit of the derivatives at 0.
- e^{-1/x^2} extended by 0 is infinitely differentiable at 0 with every Taylor coefficient zero, so its Taylor series converges and not to it.

One last question

Which single arrow carries the most of this chapter? One quantifier separates the two kinds of convergence, and most of the chapter follows from where it lands, so say whether your answer is the arrow that moves it.